Madhumitha R

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.76
Vyasa Vidyalaya Matriculation Higher Secondary School	Class XII (Tamil Nadu State Board (HSC))	2020	76.00%
Vyasa Vidyalaya Matriculation Higher Secondary School	Class X (Tamil Nadu State Board (SSLC))	2020	90.60%

ACADEMIC PROJECTS

Hyperspectral Raman Imaging and AI Integrated Segmentation for Precision Diagnostics (M.Tech Thesis) - IIT Hyderabad | Present

- **Developing** a computational pipeline for **hyperspectral Raman imaging** through spectral preprocessing, irradiance correction, dimensionality reduction, and spectral-spatial analysis, **enabling structured processing of Raman imaging data**.
- **Implementing PCA-based SLIC superpixel segmentation** to characterize spatially structured Raman regions and extract region-wise spectral representations, **supporting spectral-spatial characterization of the sample**.
- **Exploring AI-integrated segmentation approaches** for Raman spectral images, toward **automated spatial characterization and precision-diagnostic applications in biomedical research workflows.**

Non-Destructive Fruit Quality Evaluation Using VIS-NIR Spectroscopy - Anna University | IIT Madras | 2024

- **Developed** a non-destructive fruit quality assessment workflow using **portable VIS-NIR spectroscopy** and spectral-Brix data, reducing reliance on destructive quality evaluation.
- **Applied K-Means clustering, PCA, and feature selection** to identify spectral patterns **linked to soluble solids content**.
- **Built** a **PLSR model** using selected spectral features to predict fruit SSC, establishing a **computational framework** for non-destructive quality assessment from Visible Region Near-Infrared Region spectral measurements.

TECHNICAL PROJECTS

BioR&D Radar: Biomedical R&D Intelligence | Python, Streamlit, SQLite, PubMed, Sentence Transformers

- Built a literature-exploration tool to retrieve PubMed papers by topic and date range, store them in SQLite, and organize publication metadata for downstream analysis.
- Implemented publication-trend analysis using SQL, TF-IDF keyword extraction, and semantic paper connections using embeddings and cosine similarity.
- Developed rule-based screening for Raman-imaging papers and a Streamlit dashboard featuring Qwen3 8B (Ollama) for source-cited research analysis.

BioLit-RAG: Biomedical Literature Retrieval | Python, RAG, Sentence Transformers, FAISS, Qwen3 8B, Streamlit

- **Developed** a biomedical RAG application that retrieves relevant PubMed literature and generates grounded answers using a local Qwen3 8B language model.
- **Implemented** semantic retrieval using Sentence Transformers embeddings and **FAISS vector search**, ranking relevant biomedical papers before constructing LLM context.
- **Built** an interactive **Streamlit** interface displaying generated answers, similarity scores, retrieved papers, abstracts, PMIDs, and PubMed links for source-grounded literature exploration.

Tree Recommendation System | Python, Scikit-learn, Random Forest, QGIS, GeoPandas

- **Developed** a **Random Forest-based recommendation model** to predict tree suitability from environmental and spatial attributes.
- **Integrated QGIS and GeoPandas** for geospatial analysis and visualization of recommended tree species across mapped areas.
- **Automated** tree selection using **environmental and maintenance parameters**, enabling location-specific species recommendations.

A Little Better | React, TypeScript, Dexie, Supabase, PostgreSQL

- Developed a **local-first personal care PWA** with daily check-ins, journaling, weekly planning, reflection tools, and personal activity tracking for **structured self-management, motivation and progress monitoring**.
- Implemented **IndexedDB persistence through Dexie and a repository-based data layer**, separating database operations from feature-level UI components for **modular data management and independent feature development**.
- Integrated **Supabase and PostgreSQL** for authentication, cloud data synchronization, and persistent cross-device storage.

INTERNSHIPS

Intern - Bio Tech Core Instrumentation Facility (BTCIF), TICEL BIOPARK, Chennai | Jun-Jul 2023

- **Explored antimicrobial coating development** using silver and copper nanoparticles, including formulation, and MIC analysis.
- **Observed upstream and downstream bioprocessing** including fermentation, sterilization, product extraction, membrane separation, and purification workflows.
- **Gained exposure to analytical and bioprocess instrumentation** including UHPLC, GC, and elemental analysis techniques.

TECHNICAL SKILLS & RELEVANT COURSEWORK

- **Programming & Data:** Python, NumPy, Pandas, Scikit-learn, Jupyter
- **Generative AI & Retrieval:** RAG, Sentence Transformers, FAISS, Qwen3 8B
- **Machine Learning:** PCA, K-Means, Random Forest, PLSR, Feature Selection
- **Imaging & Spectroscopy:** Spectral Preprocessing, Image Processing, Image Segmentation, OpenCV
- **Software Development:**React, React Native, TypeScript, FastAPI, REST APIs, Streamlit
- **Databases:** MongoDB, PostgreSQL, Supabase, IndexedDB/Dexie

POSITIONS OF RESPONSIBILITY

1. **Coordinator - Cardiovascular Research Convergence (CRC 2026)** | IIT Hyderabad | Sep 2026
2. **Coordinator - Pre-Summit Workshop, India AI Impact Summit 2026** | IIT Hyderabad | Feb 2026
3. **PG Mentor - Sunshine Club** | IIT Hyderabad | 2026-Present
4. **PG Buddy - Sunshine Club** | IIT Hyderabad | 2025-26
5. **NSS Volunteer** | Anna University | 2020-24